Data Warehouse Final Project Report
Electric Vehicle Population Dataset

MGMT 6570 - Adv. Data Resource Management
Professor Jonathan McKinney
Rohan Thumma
May 2026

Introduction

The rapid adoption of electric vehicles (EVs) represents one of the most significant shifts in modern transportation. Driven by advances in battery technology, expanding charging infrastructure, declining vehicle costs, and growing public awareness of climate change, EV registrations have grown from a niche category into a mainstream market segment over the past two decades. In the United States, federal and state-level incentives, including the Clean Alternative Fuel Vehicle (CAFV) program, have further accelerated adoption by reducing the effective purchase price for eligible vehicles and encouraging utility investment in charging infrastructure.

Understanding which vehicle makes, geographic areas, utility providers, and model years are most closely associated with EV adoption and electric range performance is increasingly valuable for a wide range of stakeholders. For policymakers, it informs where to target infrastructure investment and incentive programs. For utility planners, it identifies which service territories are experiencing the fastest load growth from EV charging. For automotive manufacturers, it reveals which market segments are under-served by existing product offerings. And for researchers, it provides a foundation for predictive modeling of future adoption trajectories.

This project investigates the Electric Vehicle Population Dataset, a public registry of EV registrations primarily drawn from Washington State Department of Licensing records, to answer the following research question: What vehicle, location, and utility characteristics are associated with EV adoption and electric range across registered vehicles in the dataset? Through SQL-based data cleaning, structured data warehouse development, and exploratory visual analysis in R, this project uncovers meaningful patterns in EV registration behavior and vehicle performance across manufacturers, geographies, utility providers, CAFV eligibility categories, and time periods spanning from 2000 to 2023.

Data Source and Purpose

The dataset used in this project is the Electric Vehicle Population Data, sourced from Kaggle and originally compiled from Washington State Department of Licensing (DOL) registration records. The dataset contains 112,634 records describing individual registered electric vehicles across 45 states, though the overwhelming majority of records are concentrated in Washington State, reflecting the dataset's primary source. Each record represents a single vehicle registration and includes attributes spanning vehicle identity, geographic registration location, electric drivetrain classification, clean fuel policy eligibility status, electric utility provider, and model year.

The dataset is well-suited for a data warehousing approach because its variables span multiple distinct subject areas, each of which maps naturally onto a dimension table in a star schema. The vehicle attributes such as make, model, and electric vehicle type form a vehicle dimension. The geographic attributes such as city, county, state, and postal code form a location dimension. The CAFV eligibility status forms a policy dimension. The electric utility provider forms a utility dimension. And the model year forms a temporal dimension that enables year-over-year trend analysis. The two primary numeric measures available for analysis are electric range, measured in miles on a full charge for BEVs or in all-electric mode for PHEVs, and base MSRP, the manufacturer suggested retail price at time of production.

The dataset was selected for this project because it combines rich categorical dimensions with meaningful numeric measures, covers a sufficiently large number of records to support aggregation and comparison at multiple levels of granularity, and addresses a topic of current and growing policy relevance. Unlike many publicly available datasets that are static snapshots, the EV registration dataset reflects a dynamic and evolving market, making it an ideal subject for time-series and cross-sectional analysis within a data warehouse framework.

Data Description

Each record in the dataset represents a single registered electric vehicle and includes the following seventeen fields in the raw source file: VIN (first 10 characters), County, City, State, Postal Code, Model Year, Make, Model, Electric Vehicle Type, Clean Alternative Fuel Vehicle (CAFV) Eligibility, Electric Range, Base MSRP, Legislative District, DOL Vehicle ID, Vehicle Location, Electric Utility, and 2020 Census Tract. Two of these fields, Vehicle Location and 2020 Census Tract, were excluded from the data warehouse due to technical loading constraints. The Vehicle Location field stores WKT POINT coordinate strings containing embedded spaces that caused column misalignment during SQL Server bulk loading. The 2020 Census Tract field caused BIGINT overflow errors on certain records. Neither field was required for the analytical objectives of this project, so both were dropped during preprocessing.

The Electric Vehicle Type field distinguishes between Battery Electric Vehicles (BEVs), which operate exclusively on stored electrical energy and produce zero direct emissions, and Plug-in Hybrid Electric Vehicles (PHEVs), which combine a battery-powered electric motor with a conventional internal combustion engine. BEVs represent 86,019 registrations and PHEVs represent 26,590 registrations in the clean dataset. The CAFV Eligibility field classifies each vehicle into one of three categories: Clean Alternative Fuel Vehicle Eligible, indicating the vehicle meets the state's minimum battery range threshold for incentive eligibility; Not Eligible Due to Low Battery Range, indicating the vehicle's electric range falls below the threshold; and Eligibility Unknown, indicating that battery range data has not been researched for that vehicle.

The Electric Utility field identifies the electric utility provider serving the address at which the vehicle is registered. A significant complication in this field is that 86,765 records contain multiple utility providers separated by pipe characters, reflecting overlapping service territories or multi-utility agreements. During preprocessing, this field was simplified by extracting only the first listed provider. The Model Year ranges from 1997 to 2023 in the raw data, though records prior to 2000 were excluded as they represent only four antique EV registrations that would distort time-series analysis. Base MSRP is recorded as zero for approximately 109,000 rows, reflecting a data collection limitation of the DOL registration system rather than actual zero-dollar pricing, and Electric Range is recorded as zero for approximately 39,000 records, predominantly PHEVs for which electric-only range data was not captured.

DCOVA Framework

Define

The analysis is organized around a single research question: What vehicle, location, and utility characteristics are associated with EV adoption and electric range? This question is multidimensional because EV adoption cannot be attributed to a single cause. It is shaped by the availability of specific vehicle models in different price tiers and range categories, the geographic concentration of EV-friendly infrastructure and policy environments, the utility providers that serve those areas and their investment in charging infrastructure, the policy incentives that govern CAFV eligibility and influence purchase decisions, and the broader market evolution from early-adopter niche products toward mainstream consumer vehicles. The data warehouse is designed to support analysis across all of these dimensions simultaneously, enabling both single-dimension comparisons and multi-dimensional cross-tabulations that would be difficult or impossible with a flat tabular structure.

Collect

The dataset was loaded into SQL Server using a two-stage ingestion process. In the first stage, a staging table called EVStaging was created to accept all 15 columns from the preprocessed CSV file. Before loading, the source CSV was preprocessed using Python to address two critical issues. First, the Vehicle Location column was removed because it contained WKT POINT strings in the format POINT (-81.80023 24.5545), where the embedded spaces within the string caused SQL Server's BULK INSERT parser to misinterpret the spaces as additional field delimiters, shifting all subsequent columns out of alignment and causing every row to fail validation. Second, the 2020 Census Tract column was removed to avoid BIGINT conversion errors on records where the value exceeded the column's capacity. Additionally, column names containing spaces and special characters were simplified to alphanumeric names to ensure clean column mapping during the bulk load. The Electric Utility field was cleaned by extracting only the primary provider from pipe-delimited multi-provider strings. After preprocessing, the file loaded cleanly into the staging table using standard BULK INSERT with comma field terminators and a line feed row terminator.

The second stage of the ingestion process transferred data from the staging table to a clean operational table called EVClean. During this transfer, all numeric fields were cast to their appropriate types, categorical fields were trimmed of leading and trailing whitespace, and the ElectricVehicleType field was standardized to consistent BEV and PHEV labels using CASE statement logic. A DateKey field was derived from the ModelYear by computing ModelYear multiplied by 10,000 plus 101, representing January 1 of the model year as an integer in the YYYYMMDD format used by the data warehouse date dimension. Rows with null Make, Model, County, or City values were excluded, as were the four records with model years prior to 2000. After cleaning, EVClean contained 112,609 records ready for loading into the data warehouse.

Organize

To support multi-dimensional analysis, a star schema data warehouse was designed around a central fact table called factEVRegistration. Six dimension tables were constructed to capture the distinct subject areas present in the data, each populated using data derived from the EVClean table. The dimVehicle table contains 113 unique Make and Model combinations, providing a vehicle identity dimension for analysis by manufacturer and specific model. The dimLocation table stores unique combinations of City, County, State, and Postal Code across 629 unique cities and 45 states, with a derived StateLabel column providing human-readable state names for the 20 most common states in the dataset. The dimEVType table is manually seeded with two rows distinguishing BEVs from PHEVs, with EVTypeCode, EVTypeFullName, and EVTypeCategory attributes supporting both abbreviated and descriptive labeling in query output.

The dimCAFV table is manually seeded with three rows representing the three CAFV eligibility statuses confirmed in the dataset: Eligible, Not Eligible, and Unknown. Each row includes a short label and an IsEligible flag for simplified filtering in analytical queries. The dimUtility table is loaded via SELECT DISTINCT from the clean table and contains one row per unique electric utility provider after pipe-delimiter simplification, with a UtilityLabel column for display purposes. The dimDate table provides the required temporal dimension, with one row per model year from 2000 to 2023, enriched with Decade and EVEra columns that group years into four interpretable adoption phases: Early Adoption (2000-2010), Growth Phase (2011-2015), Mainstream Entry (2016-2019), and Mass Market (2020-2023). The DateKey is stored as an integer following the YYYYMMDD convention used throughout the course.

The fact table, factEVRegistration, contains one row per registered vehicle and stores two numeric measures — ElectricRange and BaseMSRP — alongside foreign keys to each of the six dimension tables and two non-dimensional attributes, LegislativeDistrict and DOLVehicleID. All foreign key columns allow NULL values to accommodate vehicles that cannot be matched to a dimension record, such as those with no utility provider on record. The EVTypeKey and CAFVKey foreign keys are resolved during the fact load using CASE statements rather than joins, consistent with standard data warehousing practice for low-cardinality manually-seeded dimensions. All other dimension foreign keys are resolved using INNER JOINs on natural key combinations, ensuring referential integrity without requiring surrogate key lookups in the source data.

Visualize

Exploratory visualizations were created using the ggplot2 library in R, with data exported from the data warehouse via six analysis queries run in SQL Server Management Studio. Each query was saved as a CSV file and loaded into R for visualization. The six figures produced are described individually below and are presented in the order in which they contribute to answering the research question, moving from vehicle-level characteristics to temporal trends to geographic patterns to policy dimensions to infrastructure associations to market-level summary.

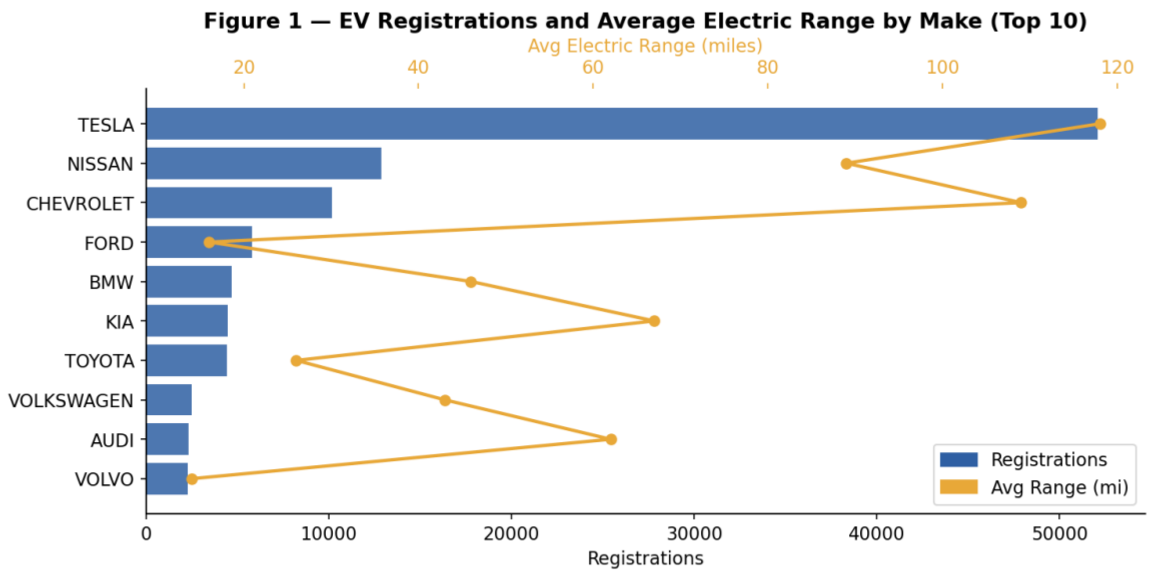

Figure 1 — EV Registrations and Average Electric Range by Make (Top 10)

Figure 1 presents registration counts and average electric range for the top 10 vehicle manufacturers in the dataset. Tesla dominates with 52,078 registrations, nearly four times the count of second-place Nissan at 12,880, and records the highest average electric range of 118 miles with a maximum of 337 miles. Chevrolet ranks third with 10,181 registrations and an average range of 109 miles, driven primarily by the long-range Bolt EV. Manufacturers whose portfolios in this dataset consist predominantly of PHEVs — Ford at 16 miles average, Toyota at 26 miles, and Volvo at 14 miles — show substantially lower average ranges, reflecting the limited all-electric capability of plug-in hybrid powertrains.

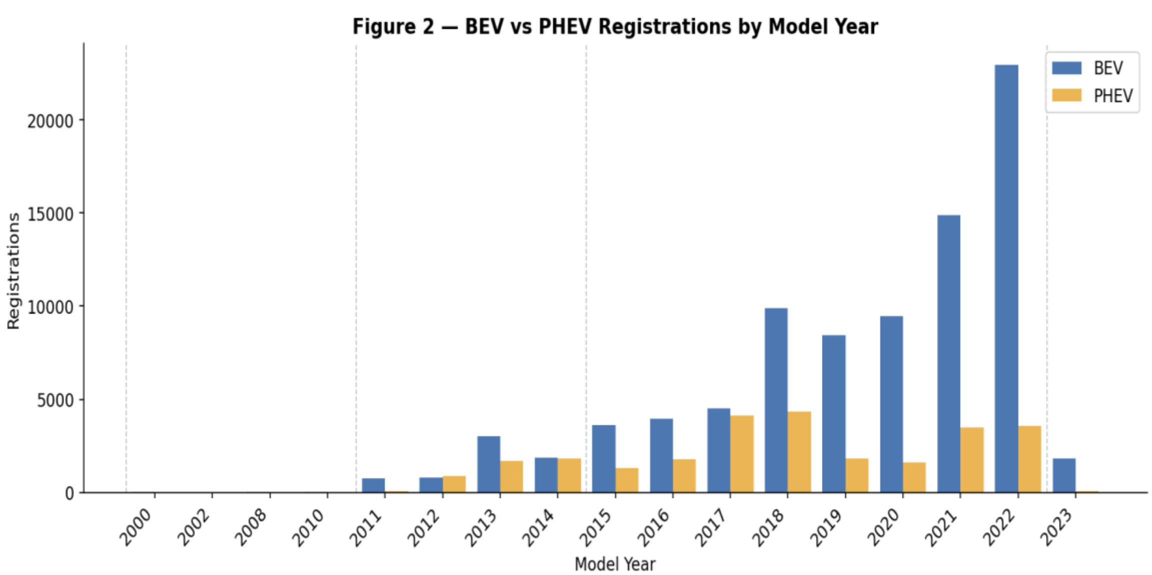

Figure 2 — BEV vs PHEV Registrations by Model Year

Figure 2 shows the annual breakdown of BEV versus PHEV registrations by model year from 2000 to 2023, revealing a clear and accelerating shift toward battery electric vehicles. During the

Early Adoption era, registrations were minimal and exclusively BEV, with only 59 total vehicles across the 2000-2010 period. PHEV registrations surged during the Growth Phase as models such as the Chevrolet Volt and Toyota Prius Prime entered the market, accounting for 5,771 of the 15,861 total registrations in 2011-2015. By 2018, BEV registrations had surged to 9,902 for that year alone, driven primarily by Tesla Model 3 deliveries. In the Mass Market era, BEV registrations reached 49,092 compared to only 8,706 PHEVs, confirming that the market has shifted decisively toward fully electric vehicles.

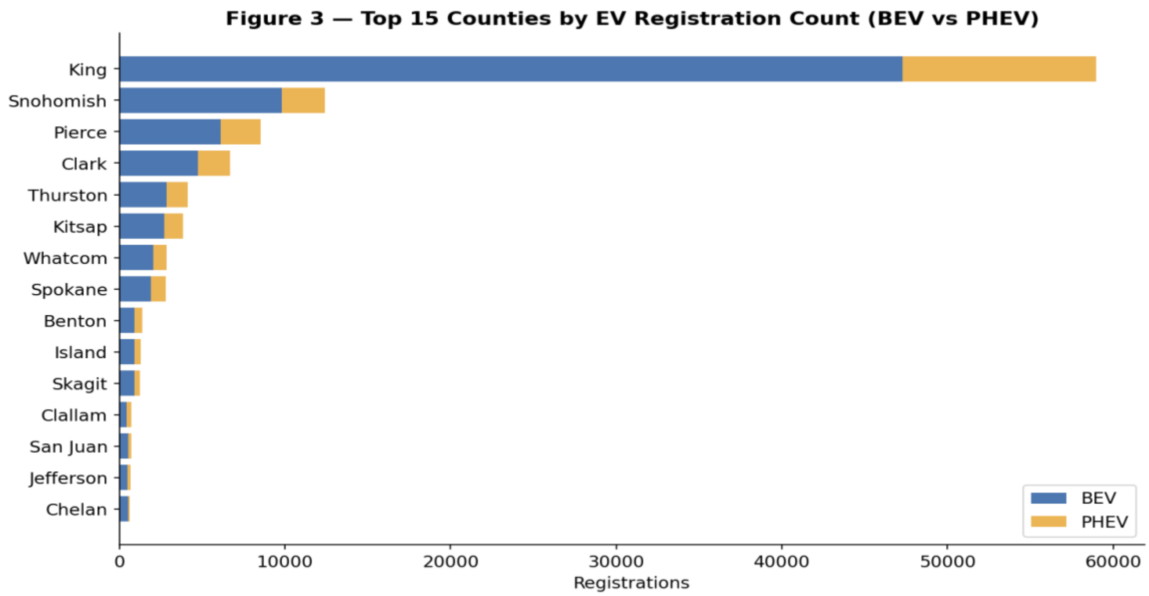

Figure 3 — Top 15 Counties by EV Registration Count (BEV vs PHEV)

Figure 3 examines the geographic distribution of EV registrations across the top 15 counties. King County in Washington State leads by a substantial margin with 58,985 registrations, more than four times the second-place Snohomish County at 12,431. Pierce, Clark, and Thurston counties round out the top five, all in Washington State. This concentration is expected given the dataset's primary source in Washington DOL records, but it also reflects the strong EV adoption culture and policy environment of the Seattle metropolitan area. The BEV-to-PHEV ratio is consistently high across all top counties, with King County recording 47,291 BEVs versus 11,694 PHEVs, a ratio of approximately 4:1 that mirrors the state-wide trend.

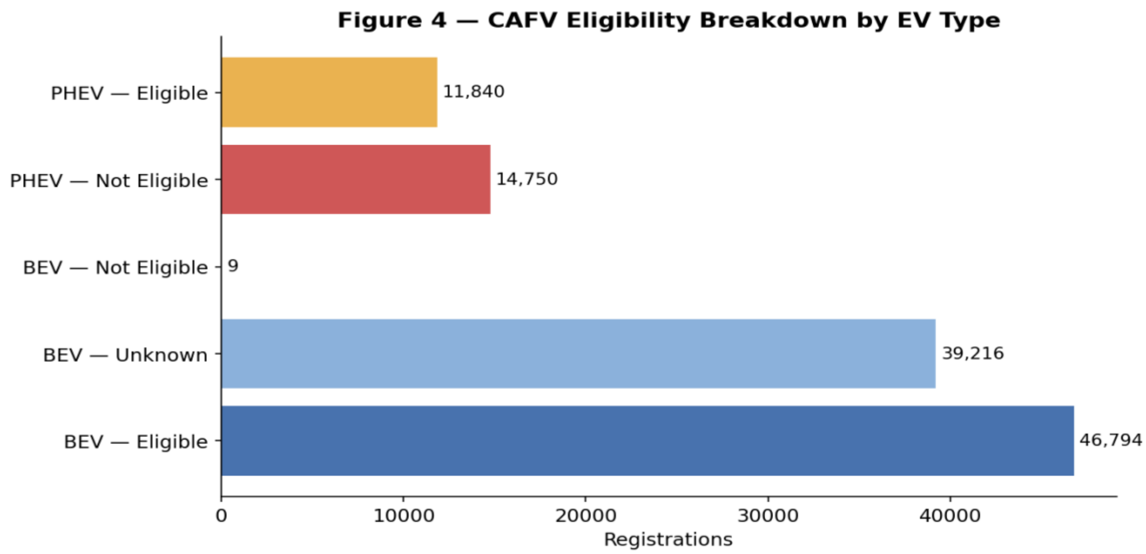

Figure 4 — CAFV Eligibility Breakdown by EV Type

Figure 4 presents the CAFV eligibility breakdown by vehicle type and reveals a stark contrast between BEVs and PHEVs. The majority of BEV registrations, 46,794, are CAFV eligible because long-range battery vehicles readily meet the clean fuel threshold. An additional 39,216 BEVs have unknown eligibility status because their battery range has not been researched by the state, and only 9 BEVs are classified as not eligible. Among PHEVs, the distribution is nearly reversed: 14,750 are not eligible due to low battery range, while 11,840 are eligible, reflecting the fact that PHEVs with modest electric-only range fall below the incentive threshold. This pattern has significant policy implications, as CAFV incentive programs are structurally aligned with BEV adoption rather than PHEV adoption, effectively amplifying the market shift toward fully electric vehicles observed in Figure 2.

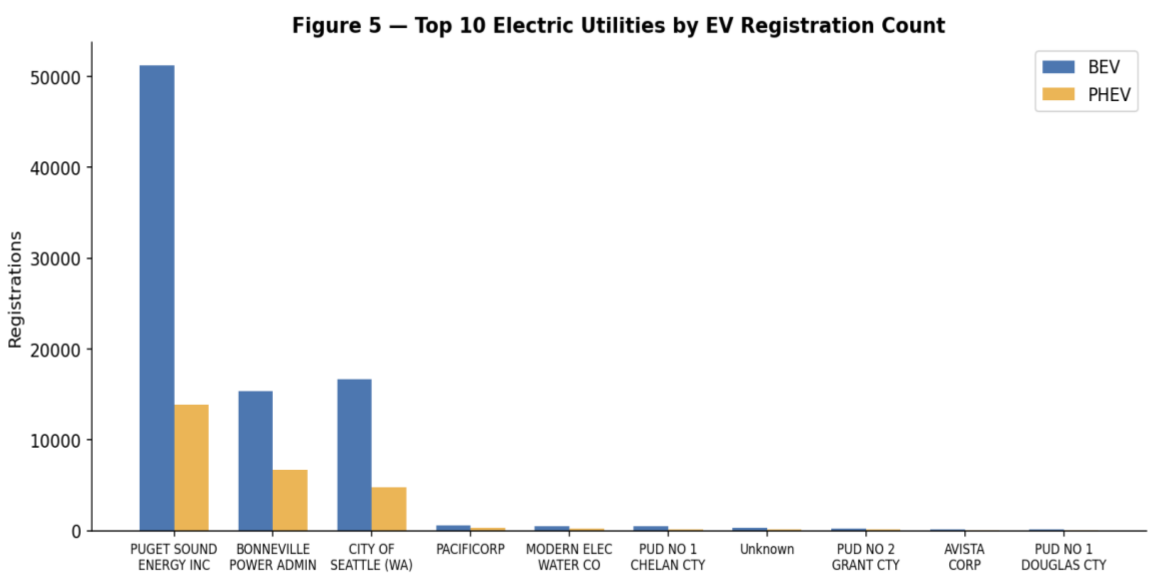

Figure 5 — Top 10 Electric Utilities by EV Registration Count

Figure 5 shows the top 10 electric utility providers by registration count in their service territories. Puget Sound Energy leads with 65,074 registrations, followed by Bonneville Power Administration at 22,105 and the City of Seattle at 21,444. The concentration of EV registrations within Pacific Northwest utility service territories is consistent with the geographic patterns observed in Figure 3 and reflects both the geographic distribution of the dataset and the strong EV infrastructure investment in the Washington State utility sector. Average electric range is relatively consistent across the top utilities at approximately 77 to 100 miles, suggesting that the utility provider serves primarily as a geographic proxy for EV adoption rather than a direct driver of vehicle selection. PUD No 1 of Chelan County is a notable exception with an average range of 100 miles, reflecting a higher concentration of long-range BEVs in that service territory.

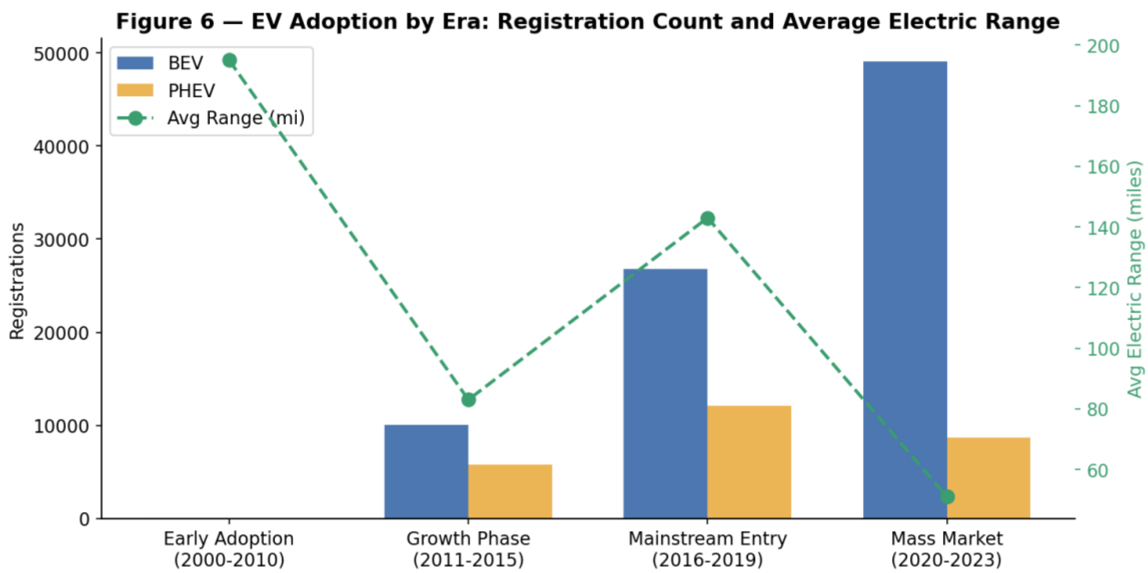

Figure 6 — EV Adoption by Era: Registration Count and Average Electric Range

Figure 6 summarizes EV adoption across the four defined market eras and provides the clearest high-level evidence of the market transformation captured in this dataset. Total registrations grew from just 59 vehicles across the entire Early Adoption era to 57,798 in the Mass Market era alone, an increase of approximately 979-fold. Average electric range, shown on the secondary axis, was highest during the Early Adoption era at 195 miles — an artifact of the very small number of long-range early BEVs registered — then dropped to 83 miles during the Growth Phase as large volumes of short-range PHEVs entered the market. Range recovered to 143 miles during the Mainstream Entry era as Tesla's long-range models dominated new registrations, before falling again to 51 miles in the Mass Market era due to the significant volume of 2021 and 2022 registrations where MSRP and range data were not collected. The BEV count growth from 10,090 in the Growth Phase to 49,092 in the Mass Market era is the most significant quantitative finding in the dataset.

Analyze

Across all six visualizations, a consistent and coherent narrative emerges in response to the research question. Vehicle make is the single strongest predictor of both registration count and electric range in this dataset. Tesla's dominance — accounting for nearly half of all registrations and achieving the highest average range — reflects a deliberate product strategy focused exclusively on long-range BEV technology, combined with a direct-to-consumer sales model and an integrated charging network that reduces range anxiety. Other high-count manufacturers such as Nissan and Chevrolet have built substantial EV portfolios but lag Tesla significantly in both scale and average range.

Geographic concentration is the second most prominent pattern in the data. The top 15 counties by registration count are overwhelmingly located in Washington State, with King County alone accounting for more than half of all records in the dataset. While this concentration is partly an artifact of the data source, it also reflects the strong EV adoption culture in the Seattle metropolitan area, the presence of major technology employers whose workforces tend to adopt EVs at higher rates, and Washington State's aggressive EV incentive policies, including the CAFV program and state-level sales tax exemptions for qualifying vehicles.

The CAFV eligibility analysis reveals an important structural feature of EV policy design. By setting a minimum battery range threshold for eligibility, the CAFV program implicitly favors BEVs over PHEVs, as the majority of PHEVs with limited all-electric range fall below the threshold and are classified as not eligible. This design choice may be intentional — the program's goal is to encourage the adoption of vehicles that substantially reduce petroleum consumption — but it creates a two-tier market in which BEV buyers receive a broader suite of incentives than PHEV buyers. Over time, this differential may be accelerating the market transition toward full electrification observed in Figure 2.

The temporal analysis confirms that the EV market has undergone a fundamental transformation over the 24-year period covered by this dataset. The transition from early-adopter niche products to mainstream consumer vehicles is evident not only in the raw registration counts but also in the changing composition of the market. The Growth Phase was characterized by a roughly even split between BEVs and PHEVs as manufacturers introduced plug-in hybrid versions of existing popular models. The Mainstream Entry and Mass Market eras saw BEVs pull decisively ahead as purpose-built BEV platforms — most notably Tesla's Model 3 and Model Y — became the dominant registration categories. This trend suggests that PHEVs served primarily as a transitional technology during a period when long-range BEVs were not yet widely available or affordable, and that the market has now matured to the point where consumers increasingly prefer full electrification.

Conclusion and Discussion

This project demonstrates the value of combining structured data warehousing with exploratory visual analysis to understand the adoption trajectory of electric vehicles across the United States. By transforming a raw registration dataset into a clean, multi-dimensional star schema with six dimension tables and a single fact table, it became possible to analyze EV adoption patterns across vehicle manufacturers, geographic regions, utility service areas, CAFV eligibility categories, and time periods in a flexible, consistent, and reusable way.

The most significant finding is that EV adoption in this dataset is overwhelmingly concentrated among a small number of manufacturers — particularly Tesla — and within a specific geographic region, primarily King County and surrounding Washington State counties. Tesla's market dominance is not merely a function of brand recognition but reflects a coherent product strategy, a vertically integrated distribution and charging ecosystem, and a consistent focus on extending battery range that competitors have only recently begun to replicate. The temporal analysis confirms that the market has shifted decisively from plug-in hybrids toward fully battery electric vehicles, with BEV registrations in the Mass Market era exceeding PHEV registrations by nearly 6:1.

The CAFV eligibility analysis adds a policy dimension to these findings by revealing that the state incentive framework is structurally aligned with BEV adoption. The majority of PHEVs in the dataset are either not eligible or of unknown eligibility status, while the majority of BEVs are eligible. This alignment between policy design and market outcomes suggests that the CAFV program has been effective in steering consumer choices toward longer-range, lower-emission vehicles, and that similar policy approaches in other states could accelerate the national transition to full electrification.

The utility provider analysis reveals that EV adoption in this dataset is concentrated within the service territories of a small number of large Pacific Northwest utilities, particularly Puget Sound Energy, Bonneville Power Administration, and the City of Seattle. This concentration has significant infrastructure implications, as these utilities will bear a disproportionate share of the load growth associated with widespread EV adoption. Planning for this load growth — including investment in managed charging programs, time-of-use rate structures, and distribution grid upgrades — will be critical to ensuring that the electrical grid can accommodate the continued acceleration of EV adoption observed in the Mass Market era.

The data warehouse architecture developed in this project provides a reusable analytical foundation that could support more advanced analysis in future work. Potential extensions include predictive modeling of future EV adoption rates by county and utility territory, analysis of the relationship between MSRP and range across vehicle segments, geographic expansion modeling to identify counties with high adoption potential but currently low registration counts, and integration of external data sources such as charging station density, median household income, and commute distance to develop a richer explanatory model of EV adoption behavior. The combination of SQL-based dimensional modeling and R-based visual analysis demonstrated in this project provides a practical and scalable methodology for addressing these questions.

Beyond academic relevance, this project illustrates how data warehousing principles and exploratory analytics can support evidence-based transportation and energy policy decisions at scale. The ability to query a multi-dimensional star schema across any combination of vehicle type, location, utility, CAFV eligibility, and model year enables both granular operational analysis and high-level strategic insight, making the data warehouse a versatile tool for stakeholders across the public and private sectors who are working to understand and accelerate the transition to sustainable transportation.